SKILL -2 : HIBERNATE CRUD OPERATIONS

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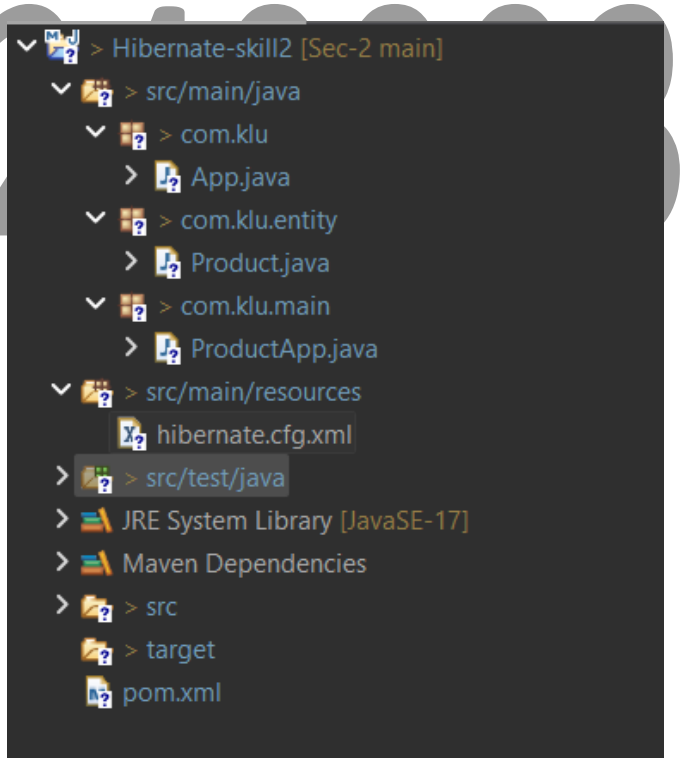
Section:2

Scenario:

A retail inventory system needs to store product details such as productId, name, description, price, and quantity. The admin should be able to add new products, retrieve product information, update price or quantity, and delete discontinued products.

Structure of File:



Codes:

Product.java:

```
package com.klu.entity;

import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.Table;

@Entity
@Table(name = "Product_skill1")
public class Product {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private int productId;   // Primary Key

    private String name;     // Product name
    private String description; // Product description
    private double price;    // Product price
    private int quantity;    // Product quantity

    // Getters and Setters
    public int getProductId() {
        return productId;
    }
    public void setProductId(int productId) {
        this.productId = productId;
    }
```

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```
public String getName() {
    return name;
}
public void setName(String name) {
    this.name = name;
}

public String getDescription() {
    return description;
}
public void setDescription(String description) {
    this.description = description;
}

public double getPrice() {
    return price;
}
public void setPrice(double price) {
    this.price = price;
}

public int getQuantity() {
    return quantity;
}
public void setQuantity(int quantity) {
    this.quantity = quantity;
}

@Override
public String toString() {
    return "Product [productId=" + productId
```

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```
            + ", name=" + name
            + ", description=" + description
            + ", price=" + price
            + ", quantity=" + quantity + "]";
    }
}
```

ProductApp.java:

```
package com.klu.main;

import org.hibernate.SessionFactory;
import org.hibernate.cfg.Configuration;
import com.klu.entity.Product;

public class ProductApp {
    public static void main(String[] args) {
        // Load Hibernate configuration
        Configuration conf = new Configuration().configure("hibernate.cfg.xml");
        SessionFactory sf = conf.buildSessionFactory();

        // 1. INSERT
        Product p = new Product();
        p.setName("Laptop");
        p.setDescription("Gaming Laptop with RTX GPU");
        p.setPrice(1200.50);
        p.setQuantity(10);

        org.hibernate.Session s = sf.openSession();
        org.hibernate.Transaction tx = s.beginTransaction();
        s.persist(p);   // Save product
        tx.commit();
```

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```
    s.close();

    // 2. READ
    s = sf.openSession();
    Product readProduct = s.find(Product.class, 1); // Replace 1 with actual productId
    System.out.println("Read: " + readProduct);
    s.close();

    // 3. UPDATE
    s = sf.openSession();
    Product updateProduct = s.find(Product.class, 1);
    updateProduct.setName("High-End Laptop");
    updateProduct.setDescription("Upgraded Gaming Laptop with RTX 5090 GPU");
    updateProduct.setPrice(1800.75);
    updateProduct.setQuantity(5);

    tx = s.beginTransaction();
    s.merge(updateProduct);  // Update product
    tx.commit();
    s.close();

    // 4. DELETE
    s = sf.openSession();
    Product deleteProduct = s.find(Product.class, 1);
    tx = s.beginTransaction();
    s.remove(deleteProduct); // Delete product
    tx.commit();
    s.close();

    sf.close();
}
```

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}

Hibernate.cfg.xml:

```
<!DOCTYPE hibernate-configuration PUBLIC
 "-//Hibernate/Hibernate Configuration DTD 3.0//EN"
 "http://www.hibernate.org/dtd/hibernate-configuration-3.0.dtd">

<hibernate-configuration>
 <session-factory>
  <!-- DB properties -->
  <property name="hibernate.connection.driver_class">
   com.mysql.cj.jdbc.Driver
  </property>
  <property name="hibernate.connection.url">
   jdbc:mysql://localhost:3306/sec2
  </property>
  <property name="hibernate.connection.username">
   root
  </property>
  <property name="hibernate.connection.password">
   #Sudheer@2007
  </property>

  <!-- Hibernate properties -->
  <property name="hibernate.dialect">
      org.hibernate.dialect.MySQL8Dialect
  </property>
  <property name="hibernate.hbm2ddl.auto">
    update
  </property>
  <property name="hibernate.show_sql">
```

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```
    true
   </property>

   <!-- mapping class information -->
   <mapping class="com.klu.entity.Product"/>
  </session-factory>
</hibernate-configuration>
```

pom.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
        <modelVersion>4.0.0</modelVersion>

        <groupId>com.klu</groupId>
        <artifactId>HibernateCrudApp</artifactId>
        <version>0.0.1-SNAPSHOT</version>

        <name>HibernateCrudApp</name>
        <!-- FIXME change it to the project's website -->
        <url>http://www.example.com</url>

        <properties>
                <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
                <maven.compiler.release>17</maven.compiler.release>
        </properties>

        <dependencyManagement>
                <dependencies>
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```
                <dependency>
                    <groupId>org.junit</groupId>
                    <artifactId>junit-bom</artifactId>
                    <version>5.11.0</version>
                    <type>pom</type>
                    <scope>import</scope>
                </dependency>
            </dependencies>
        </dependencyManagement>

        <dependencies>
        <!-- Hibernate Core 6 -->
<dependency>
  <groupId>org.hibernate.orm</groupId>
  <artifactId>hibernate-core</artifactId>
  <version>6.4.4.Final</version>
</dependency>


<!-- MySQL Connector -->
<dependency>
  <groupId>mysql</groupId>
  <artifactId>mysql-connector-java</artifactId>
  <version>8.0.33</version>
</dependency>

<!-- Jakarta Persistence API -->
<dependency>
  <groupId>jakarta.persistence</groupId>
  <artifactId>jakarta.persistence-api</artifactId>
  <version>3.1.0</version>
```

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```
</dependency>
        <!-- https://mvnrepository.com/artifact/com.mysql/mysql-connector-j -->
        <dependency>
            <groupId>com.mysql</groupId>
            <artifactId>mysql-connector-j</artifactId>
            <version>8.0.33</version>
        </dependency>
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter-api</artifactId>
            <scope>test</scope>
        </dependency>
        <!-- Optionally: parameterized tests support -->
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter-params</artifactId>
            <scope>test</scope>
        </dependency>
    </dependencies>

    <build>
        <pluginManagement><!-- lock down plugins versions to avoid using Maven
            defaults (may be moved to parent pom) -->
            <plugins>
                <!-- clean lifecycle, see
                https://maven.apache.org/ref/current/maven-core/lifecycles.html#clean_Lifecycle -->
                <plugin>
                    <artifactId>maven-clean-plugin</artifactId>
                    <version>3.4.0</version>
                </plugin>
```

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```
<!-- default lifecycle, jar packaging: see
https://maven.apache.org/ref/current/maven-core/default-bindings.html#Plugin_bindings_for_jar_packaging -->
<plugin>
    <artifactId>maven-resources-plugin</artifactId>
    <version>3.3.1</version>
</plugin>
<plugin>
    <artifactId>maven-compiler-plugin</artifactId>
    <version>3.13.0</version>
</plugin>
<plugin>
    <artifactId>maven-surefire-plugin</artifactId>
    <version>3.3.0</version>
</plugin>
<plugin>
    <artifactId>maven-jar-plugin</artifactId>
    <version>3.4.2</version>
</plugin>
<plugin>
    <artifactId>maven-install-plugin</artifactId>
    <version>3.1.2</version>
</plugin>
<plugin>
    <artifactId>maven-deploy-plugin</artifactId>
    <version>3.1.2</version>
</plugin>
<!-- site lifecycle, see
https://maven.apache.org/ref/current/maven-core/lifecycles.html#site_Lifecycle -->
<plugin>
    <artifactId>maven-site-plugin</artifactId>
```

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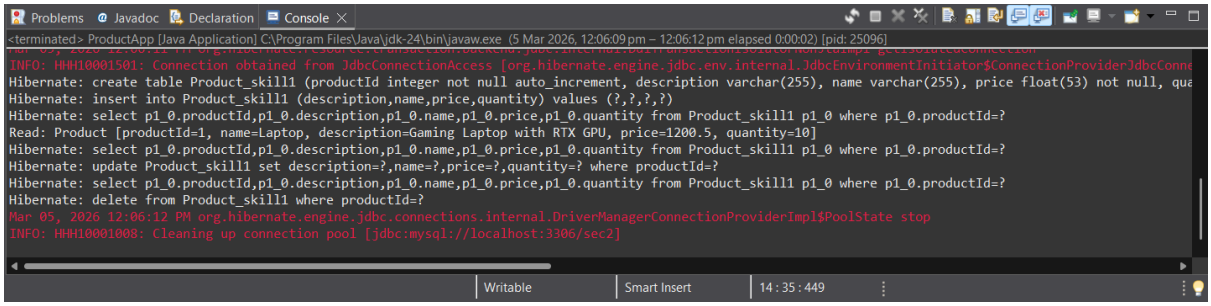
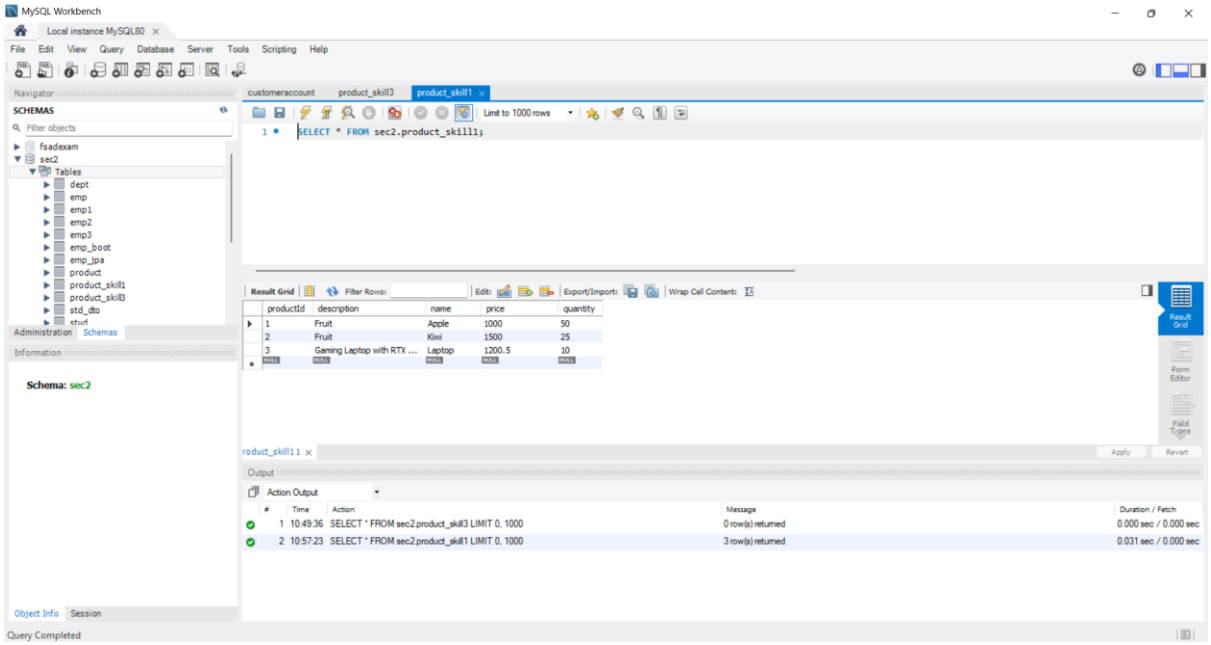
```
                <version>3.12.1</version>
            </plugin>
            <plugin>
                <artifactId>maven-project-info-reports-plugin</artifactId>
                <version>3.6.1</version>
            </plugin>
        </plugins>
    </pluginManagement>
  </build>
</project>
```

Outputs:





GITHUB LINK:

https://github.com/nagasubrahmanyammeka/SkillExperiments_2400030128_FSAD_Sec2/tree/main/SKILL-2

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